

Lecture Slide - 3

Root Finding Problems - Bisection Method

Root Finding Problems

Many problems in Science and Engineering are expressed as:

Given a continuous function $f(x)$,
find the value r such that $f(r) = 0$

These problems are called root finding problems.

Roots of Equations

A number r that satisfies an equation is called a root of the equation.

The equation: $x^4 - 3x^3 - 7x^2 + 15x = -18$

has four roots: -2 , 3 , 3 , and -1 .

i.e., $x^4 - 3x^3 - 7x^2 + 15x + 18 = (x + 2)(x - 3)^2(x + 1)$

The equation has two simple roots (-1 and -2) and a repeated root (3) with multiplicity = 2.

Zeros of a Function

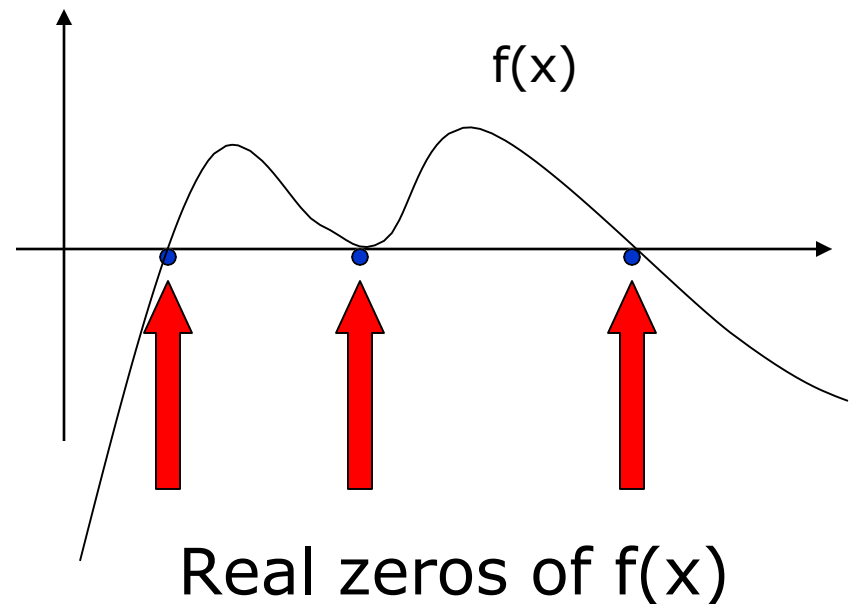
Let $f(x)$ be a real-valued function of a real variable. Any number r for which $f(r)=0$ is called a zero of the function.

Examples:

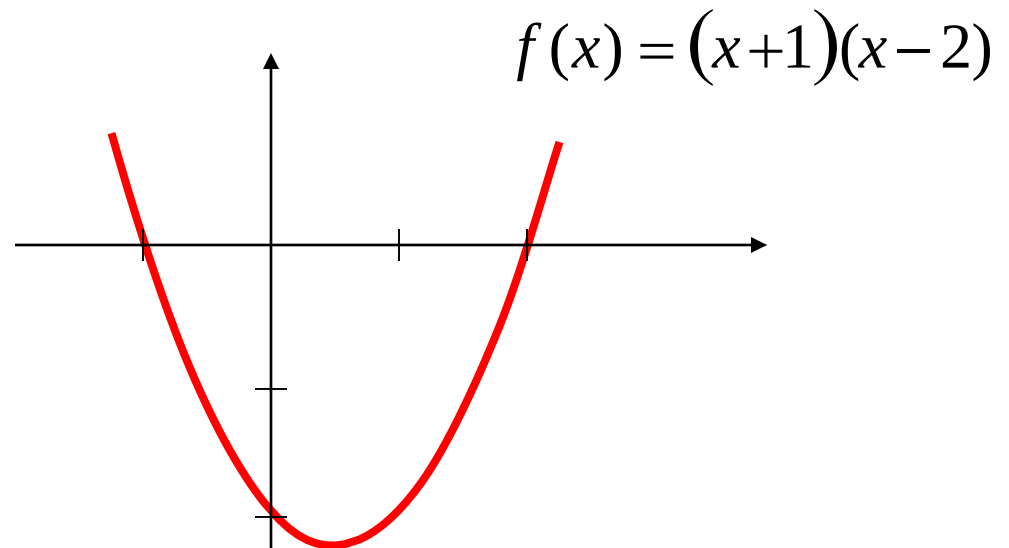
2 and 3 are zeros of the function $f(x) = (x-2)(x-3)$.

Graphical Interpretation of Zeros

- The real zeros of a function $f(x)$ are the values of x at which the graph of the function crosses (or touches) the x-axis.



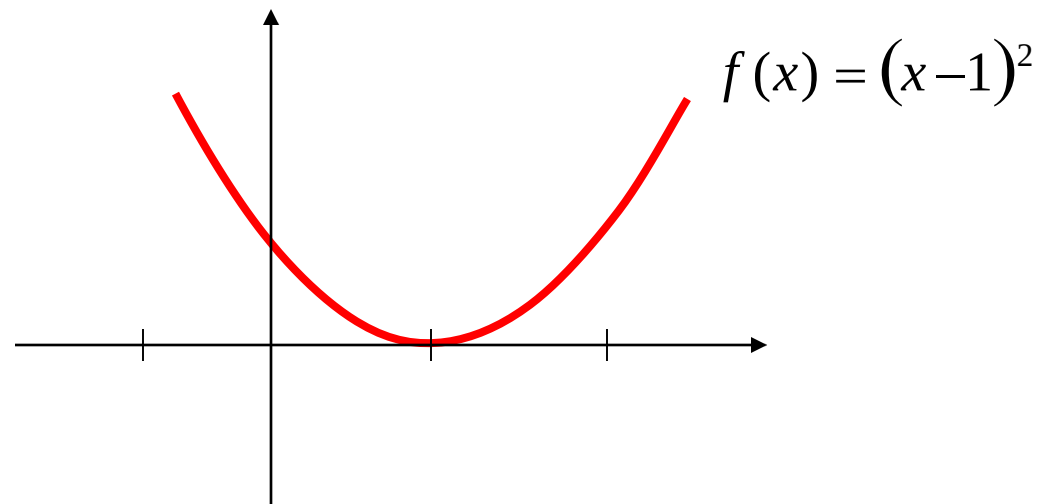
Simple Zeros



$$f(x) = (x+1)(x-2) = x^2 - x - 2$$

has two simple zeros (one at $x = 2$ and one at $x = -1$)

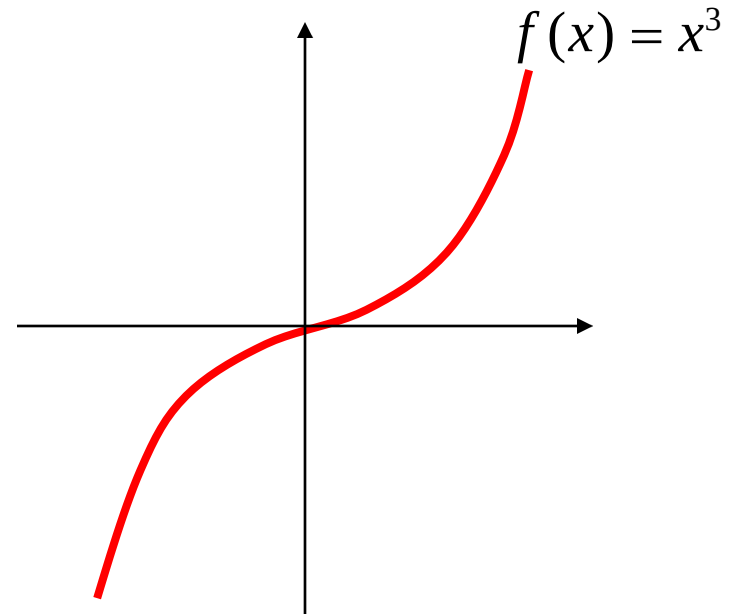
Multiple Zeros



$$f(x) = (x-1)^2 = x^2 - 2x + 1$$

has double zeros (zero with multiplicity = 2) at $x = 1$

Multiple Zeros



$$f(x) = x^3$$

has a zero with multiplicity = 3 at $x = 0$

Facts

- Any n^{th} order polynomial has exactly n zeros (counting real and complex zeros with their multiplicities).
- Any polynomial with an odd order has at least one real zero.
- If a function has a zero at $x=r$ with multiplicity m , then the function and its first $(m-1)$ derivatives are zero at $x=r$, and the m^{th} derivative at r is not zero.

Roots of Equations & Zeros of Function

Given the equation:

$$x^4 - 3x^3 - 7x^2 + 15x = -18$$

Move all terms to one side of the equation:

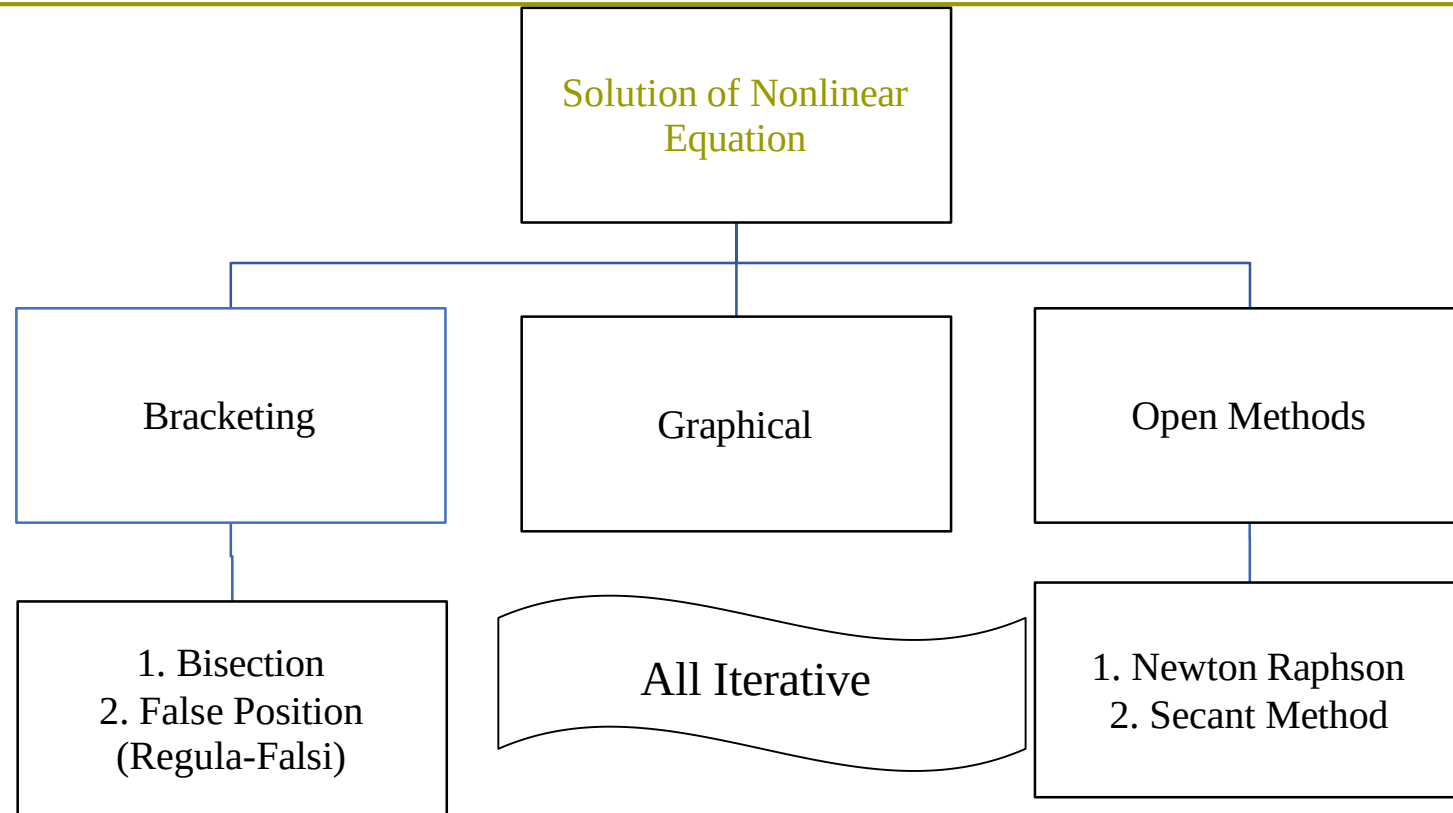
$$x^4 - 3x^3 - 7x^2 + 15x + 18 = 0$$

Define $f(x)$ as:

$$f(x) = x^4 - 3x^3 - 7x^2 + 15x + 18$$

The zeros of $f(x)$ are the same as the roots of the equation $f(x) = 0$ (Which are $-2, 3, 3$, and -1)

Solution Methods



Solution Methods

Several ways to solve nonlinear equations are possible:

- Analytical Solutions
 - Possible for special equations only
- Graphical Solutions
 - Useful for providing initial guesses for other methods
- Numerical Solutions
 - Open methods
 - Bracketing methods

Analytical Methods

Analytical Solutions are available for special equations only.

Analytical solution of : $ax^2 + bx + c = 0$

$$\text{roots} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

No analytical solution is available for: $x - e^{-x} = 0$

Graphical Methods

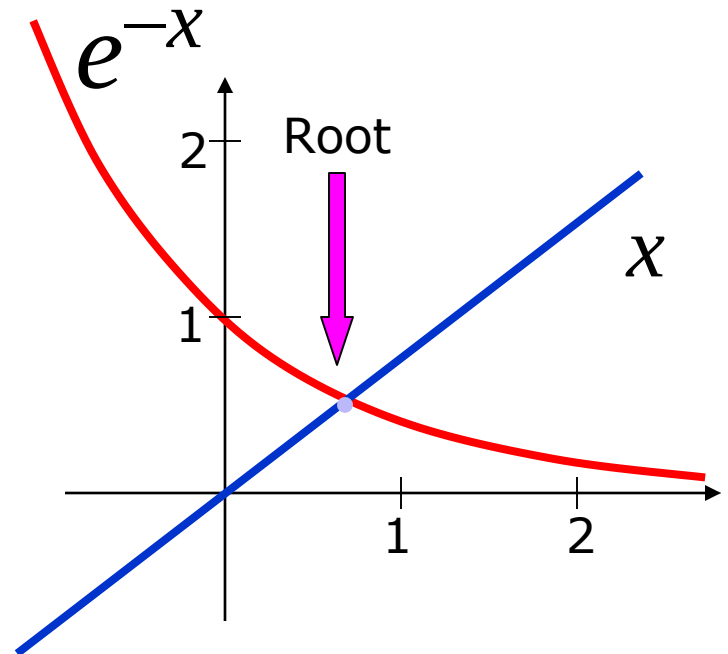
- Graphical methods are useful to provide an initial guess to be used by other methods.

Solve

$$x = e^{-x}$$

The root $\in [0,1]$

root ≈ 0.6



Bracketing Methods

- In bracketing methods, the method starts with an interval that contains the root and a procedure is used to obtain a smaller interval containing the root.
- Examples of bracketing methods:
 - Bisection method
 - False position method

Open Methods

- ❑ In the open methods, the method starts with one or more initial guess points. In each iteration, a new guess of the root is obtained.
- ❑ Open methods are usually more efficient than bracketing methods.
- ❑ They may not converge to a root.

Bisection Method



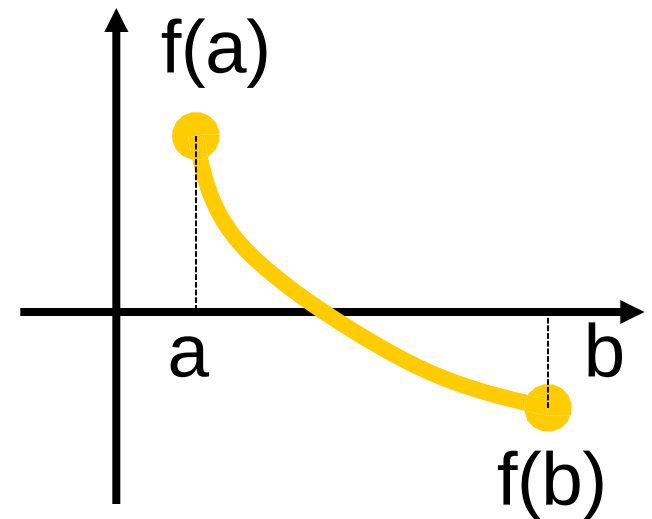
- The Bisection Algorithm
- Convergence Analysis of Bisection Method
- Examples

Introduction

- ❑ The **Bisection method** is one of the simplest methods to find a zero of a nonlinear function.
- ❑ It is also called the **interval halving** method.
- ❑ To use the Bisection method, one needs an initial interval that is known to contain a zero of the function.
- ❑ The method **systematically reduces the interval**. It does this by dividing the interval into two equal parts, performing a simple test, and based on the result of the test, half of the interval is thrown away.
- ❑ The procedure is repeated until the desired interval size is obtained.

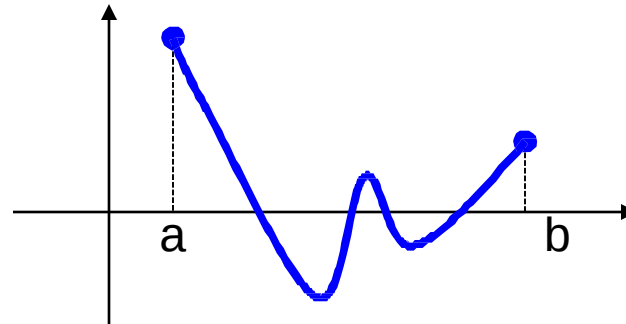
Intermediate Value Theorem

- Let $f(x)$ be defined on the interval $[a,b]$.
- Intermediate value theorem:
if a function is continuous and $f(a)$ and $f(b)$ have different signs then the function has at least one zero in the interval $[a,b]$.



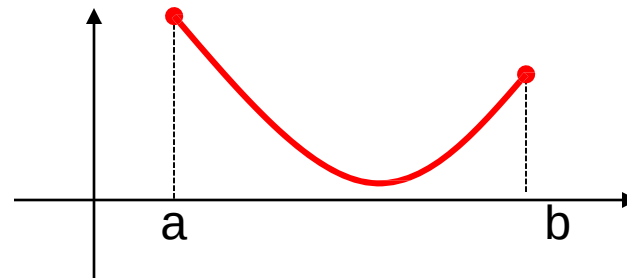
Examples

- If $f(a)$ and $f(b)$ have the same sign, the function may have an even number of real zeros or no real zeros in the interval $[a, b]$.



The function has four real zeros

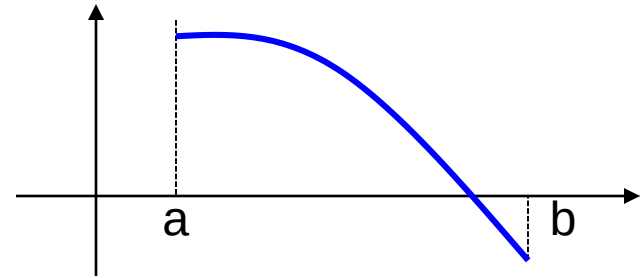
- Bisection method can not be used in these cases.



The function has no real zeros

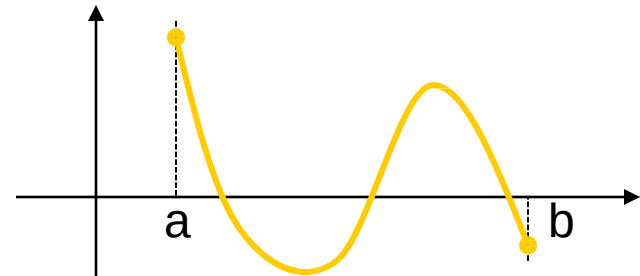
Two More Examples

- If $f(a)$ and $f(b)$ have different signs, the function has at least one real zero.



The function has one real zero

- The bisection method can be used to find one of the zeros.



The function has three real zeros

Bisection Method

- If the function is continuous on $[a,b]$ and $f(a)$ and $f(b)$ have different signs, the Bisection method obtains a new interval that is half of the current interval. The sign of the function at the endpoints of the interval is different.
- This allows us to repeat the Bisection procedure to further reduce the size of the interval.

Bisection Method

Assumptions:

Given an interval $[a,b]$

$f(x)$ is continuous on $[a,b]$

$f(a)$ and $f(b)$ have opposite signs.

These assumptions ensure the existence of at least one zero in the interval $[a,b]$ and the bisection method can be used to obtain a smaller interval that contains the zero.

Bisection Method

For the arbitrary equation of one variable, $f(x) = 0$

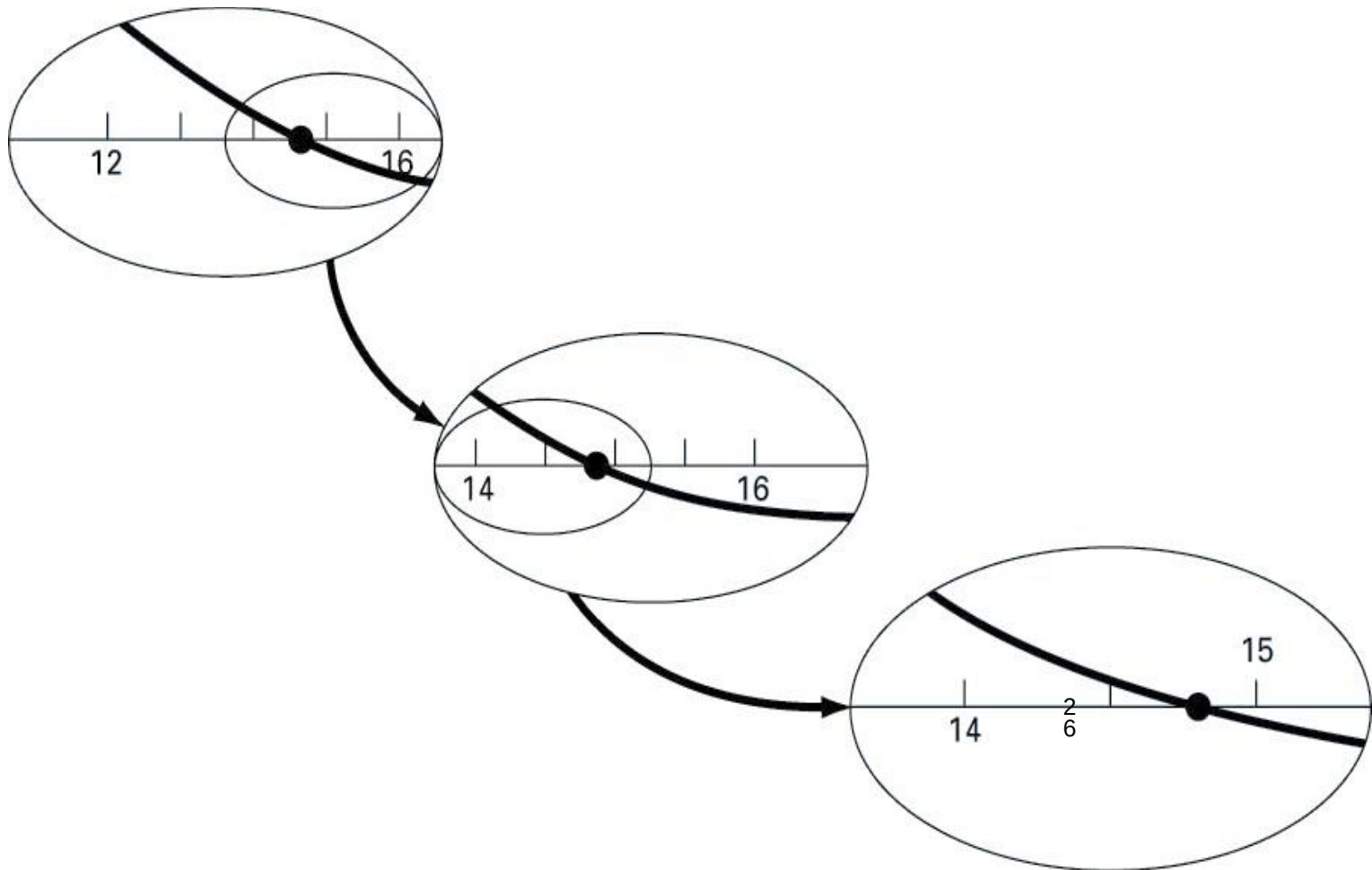
1. Pick x_l and x_u such that they bound the root of interest, and check if $f(x_l) \times f(x_u) < 0$.
2. Estimate the root by evaluating $f[(x_l+x_u)/2]$.
3. Find the pair
 - If $f(x_l) \times f[(x_l+x_u)/2] < 0$, root lies in the lower interval, then $x_u = (x_l+x_u)/2$ and go to step 2.

Bisection Method

- If $f(x_l) \times f[(x_l+x_u)/2] > 0$, root lies in the upper interval, then $x_l = [(x_l+x_u)/2]$, go to step 2.
 - If $f(x_l) \times f[(x_l+x_u)/2] = 0$, then root is $(x_l+x_u)/2$ and terminate.
- Compare ε_s with ε_a
- If $\varepsilon_a < \varepsilon_s$, stop. Otherwise repeat the process.


$$\left\{ \begin{array}{l} \frac{\left| x_l - \frac{x_l + x_u}{2} \right|}{\left| \frac{x_l + x_u}{2} \right|} \leq 100\% \\ \text{or} \\ \frac{\left| x_u - \frac{x_l + x_u}{2} \right|}{\left| \frac{x_l + x_u}{2} \right|} \leq 100\% \end{array} \right\}$$

Bisection Method



Bisection Algorithm (Summary)

Assumptions:

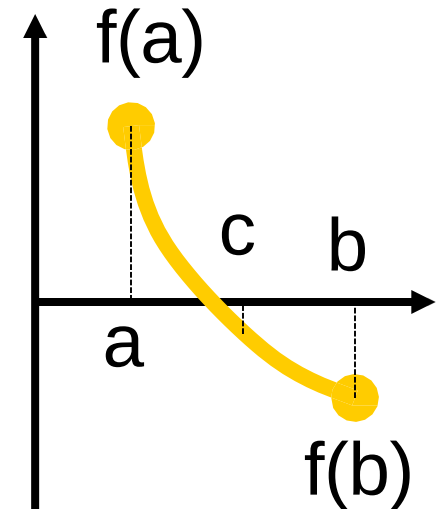
- ▣ $f(x)$ is continuous on $[a, b]$
- ▣ $f(a) f(b) < 0$

Algorithm:

Loop

1. Compute the mid point $c = (a+b)/2$
2. Evaluate $f(c)$
3. If $f(a) f(c) < 0$ then new interval $[a, c]$
If $f(a) f(c) > 0$ then new interval $[c, b]$

End loop



Example

Can you use the Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval $[0,2]$?

Answer:

$f(x)$ is continuous on $[0,2]$ and

$$f(0) \cdot f(2) = (1)(3) = 3 > 0$$

$\Rightarrow$ Assumptions are not satisfied

$\Rightarrow$ The bisection method cannot be used

Example

Can you use Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval $[0,1]$?

Answer:

$f(x)$ is continuous on $[0,1]$

and $f(0) \cdot f(1) = (1)(-1) = -1 < 0$

$\Rightarrow$ Assumptions are satisfied

$\Rightarrow$ Bisection method can be used

Stopping Criteria

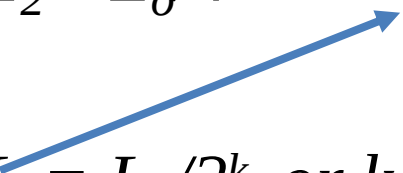
Two common stopping criteria

1. Stop after a fixed number of iterations
2. Stop when the absolute error is less than a specified value

How are these criteria related?

How Many Iterations Will It Take?

Another benefit of the bisection method is that the number of iterations required to attain an absolute error can be computed *a priori*—that is, before starting the iterations

- Length of the first Interval $L_0 = b - a$
 - After 1 iteration $L_1 = L_0/2$
 - After 2 iterations $L_2 = L_0/4$
 - ...
 - After k iterations $L_k = L_0/2^k$ or $k = \ln_2(L_0/L_k)$
-  Absolute Error

$$\epsilon_a \leq \frac{L_k}{x} \times 100\% \quad \epsilon_a \leq \epsilon_s$$

How Many Iterations Will It Take?

- If the absolute magnitude of the error is $\frac{\epsilon_s \cdot X}{100\%} = 10^{-4}$

And $L_0 = 2$, how many iterations will you have to do to get the required accuracy in the solution?

$$10^{-4} = \frac{2}{2^k} \Rightarrow 2^k = 2 \times 10^4 \Rightarrow k \cong 14.3 = 15$$

Best Estimate and Error Level

Bisection method obtains an interval that is guaranteed to contain a zero of the function.

Questions:

- What is the best estimate of the zero of $f(x)$?
- What is the error level in the obtained estimate?

Best Estimate and Error Level

The best estimate of the zero of the function **$f(\mathbf{x})$** after the first iteration of the Bisection method is the mid point of the initial interval:

$$\textit{Estimate of the zero: } r = \frac{b + a}{2}$$

$$\textit{Error} \leq \frac{b - a}{2}$$

Example

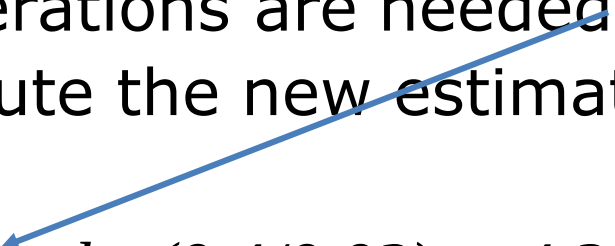
- Use Bisection method to find a root of the equation $x = \cos(x)$ with absolute error < 0.02 (assume the initial interval $[0.5, 0.9]$)

Question 1: What is $f(x)$?

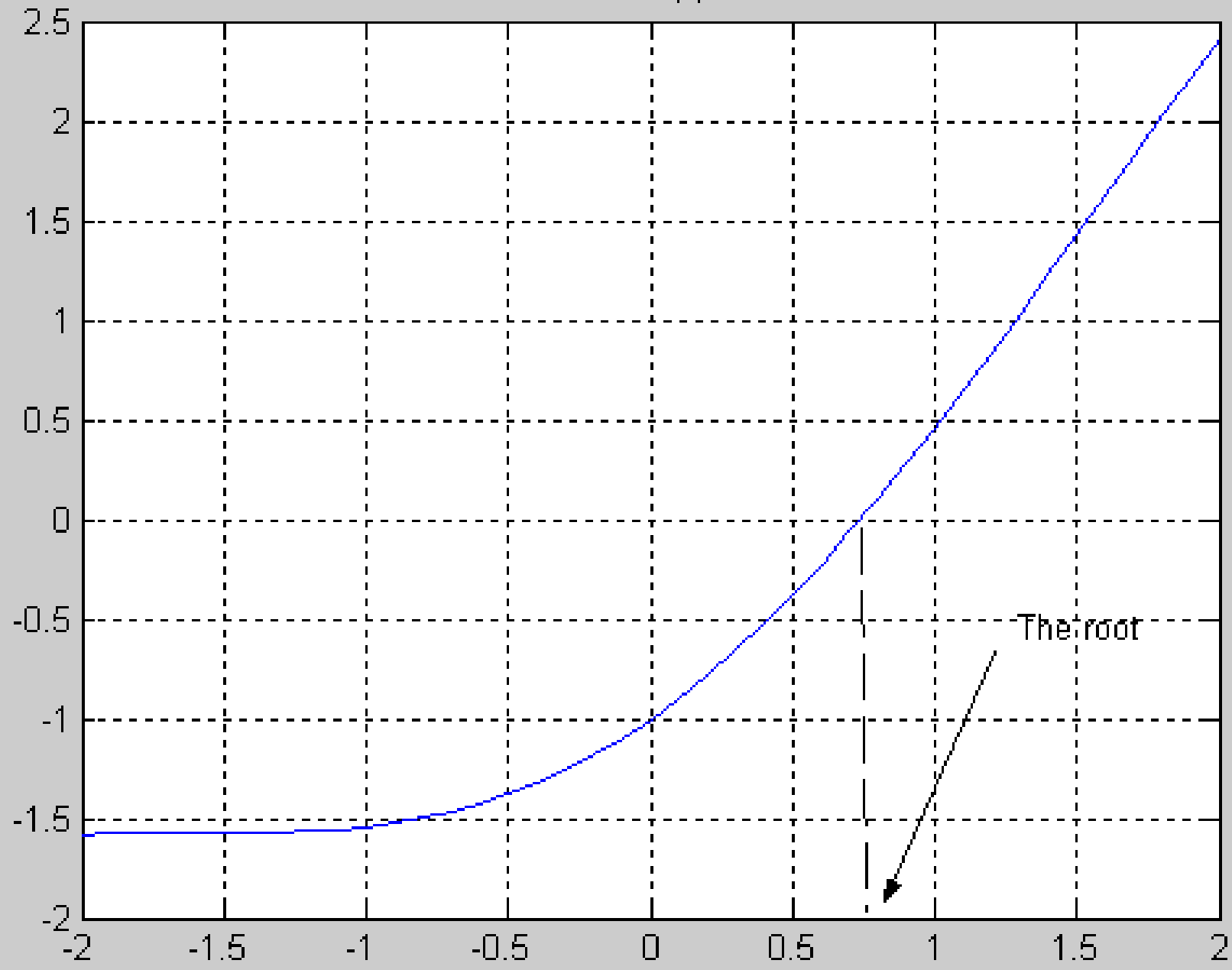
Question 2: Are the assumptions satisfied ?

Question 3: How many iterations are needed ?

Question 4: How to compute the new estimate ?


$$k = \ln_2(0.4/0.02) = 4.322 = 5$$

$$x - \cos(x)$$



Bisection Method

Initial Interval

$$f(a) = -0.3776$$

$$f(b) = 0.2784$$

Error < 0.2

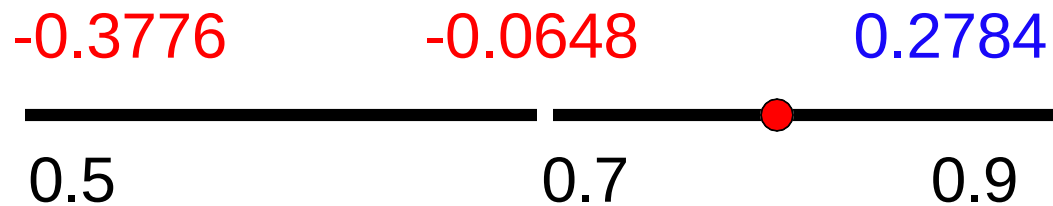


$$a = 0.5$$

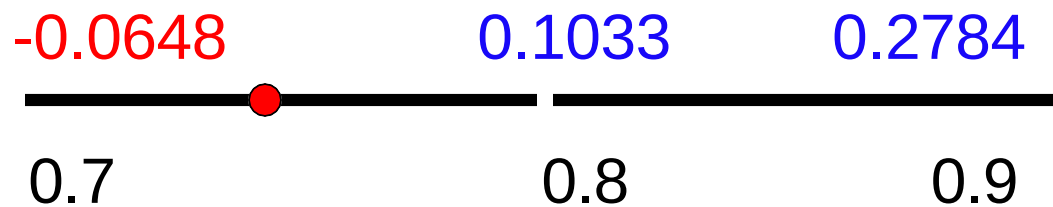
$$c = 0.7$$

$$b = 0.9$$

Bisection Method

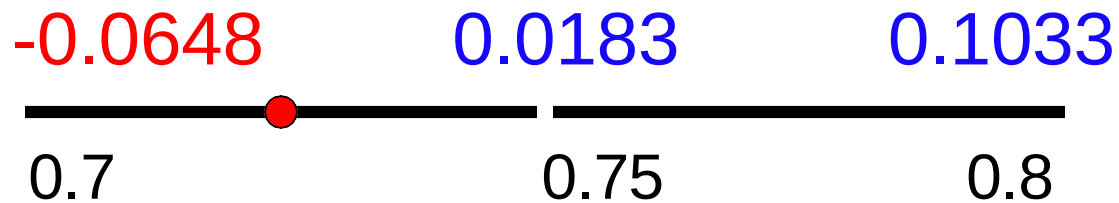


Error < 0.1

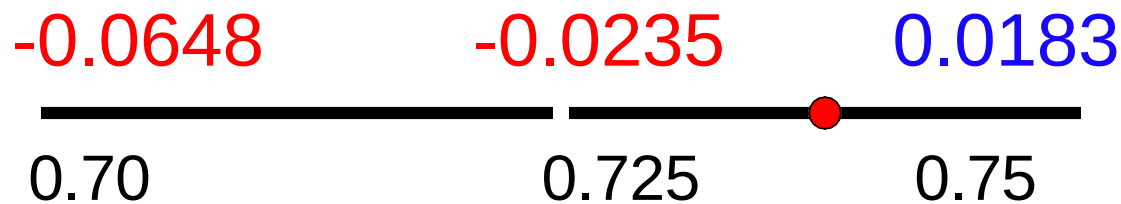


Error < 0.05

Bisection Method



Error < 0.025



Error < .0125

Summary

- ❑ Initial interval containing the root:
 $[0.5, 0.9]$

- ❑ After 5 iterations:
 - Interval containing the root: $[0.725, 0.75]$
 - Best estimate of the root is 0.7375
 - $| \text{Error} | < 0.0125$

More Example

Problem Statement. Use bisection method to locate the root of $f(x) = x^{10} - 1$ between $x = 0$ and 1.3 .

Solution. Using bisection, the results can be summarized as

$$\varepsilon_a = |(x_l - x_r)/x_r| * 100\% \quad \text{and} \quad \varepsilon_t = |(Root - x_r)/Root| * 100\%$$

Iteration	x_l	x_u	$x_r = (x_l + x_u)/2$	$\varepsilon_a (\%)$	$\varepsilon_t (\%)$
1	0	1.3	0.65	100.0	35
2	0.65	1.3	0.975	33.3	2.5
3	0.975	1.3	1.1375	14.3	13.8
4	0.975	1.1375	1.05625	7.7	5.6
5	0.975	1.05625	1.015625	4.0	1.6

Thus, after five iterations, the true error is reduced to less than 2%.

More Example

Find the root of:

$$f(x) = x^3 - 3x + 1 \text{ in the interval: } [0, 1]$$

* $f(x)$ is continuous

* $f(0) = 1, f(1) = -1 \Rightarrow f(a) f(b) < 0$

$\Rightarrow$ Bisection method can be used to find the root

More Example

Iteration	a	b	$c = \frac{(a+b)}{2}$	f(c)	$\epsilon_a (\%)$	$\epsilon_t (\%)$
1	0	1	0.5	-0.375	?	?
2	0	0.5	0.25	0.266	?	?
3	0.25	0.5	.375	-7.23E-3	?	?
4	0.25	0.375	0.3125	9.30E-2	?	?
5	0.3125	0.375	0.34375	9.37E-3	?	?

Bisection Method

Advantages

- ❑ Simple and easy to implement
- ❑ One function evaluation per iteration
- ❑ The size of the interval containing the zero is reduced by 50% after each iteration
- ❑ The number of iterations can be determined a priori
- ❑ No knowledge of the derivative is needed
- ❑ The function does not have to be differentiable

Disadvantage

- ❑ Slow to converge
- ❑ Good intermediate approximations may be discarded